



Enterprise Infrastructure Deployment

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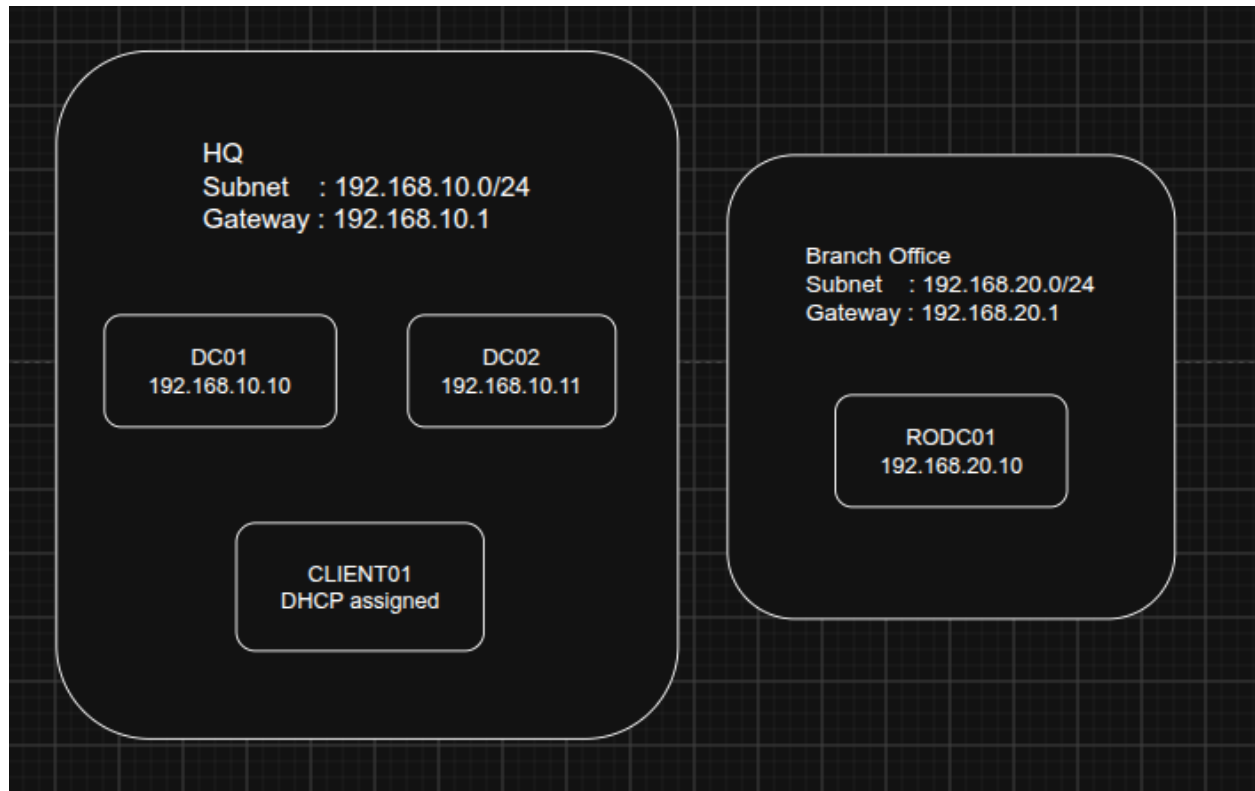
Execution Summary

The ITSimplera Solution Enterprise Infrastructure Deployment project successfully transitioned the company from a fragmented work-group model to a robust, centralized Active Directory Domain Services (AD DS) environment. Deployed via VMware ESXi, the new architecture delivers strict security enforcement across a centralized Headquarters (HQ) and a remote Branch Office. The deployment incorporates redundant domain controllers, automated network services (DHCP/DNS), localized branch caching via a Read-Only Domain Controller (RODC), and security enforced through Group Policy Objects (GPOs).

Infrastructure Overview

```
[ ITSimplera Solution Enterprise Forest: itsimplera.local ]
├── Headquarters (HQ) [192.168.10.0/24]
│   ├── DC01 (Primary DC, DNS, DHCP) [192.168.10.10]
│   ├── DC02 (Additional DC, Redundancy) [192.168.10.11]
│   ├── CLIENT01 (Domain Workstation) [Pool: 192.168.10.100-200]
│   └── Organizational Units (OUs)
│       ├── HQ-OU
│       │   ├── IT Sub-OU
│       │   ├── HR Sub-OU
│       │   └── Finance Sub-OU
│       └── Branch Office [192.168.20.0/24]
│           ├── RODC01 (Read-Only DC) [192.168.20.10]
│           └── Organizational Units (OUs)
│               └── BranchOffice-OU
```

Network Topology



IP Addressing Table

HQ

DC01	192.168.10.10 / 24
DC02	192.168.10.11 / 24
CLIENT01	DHCP assigned
Gateway	192.168.10.1

Branch

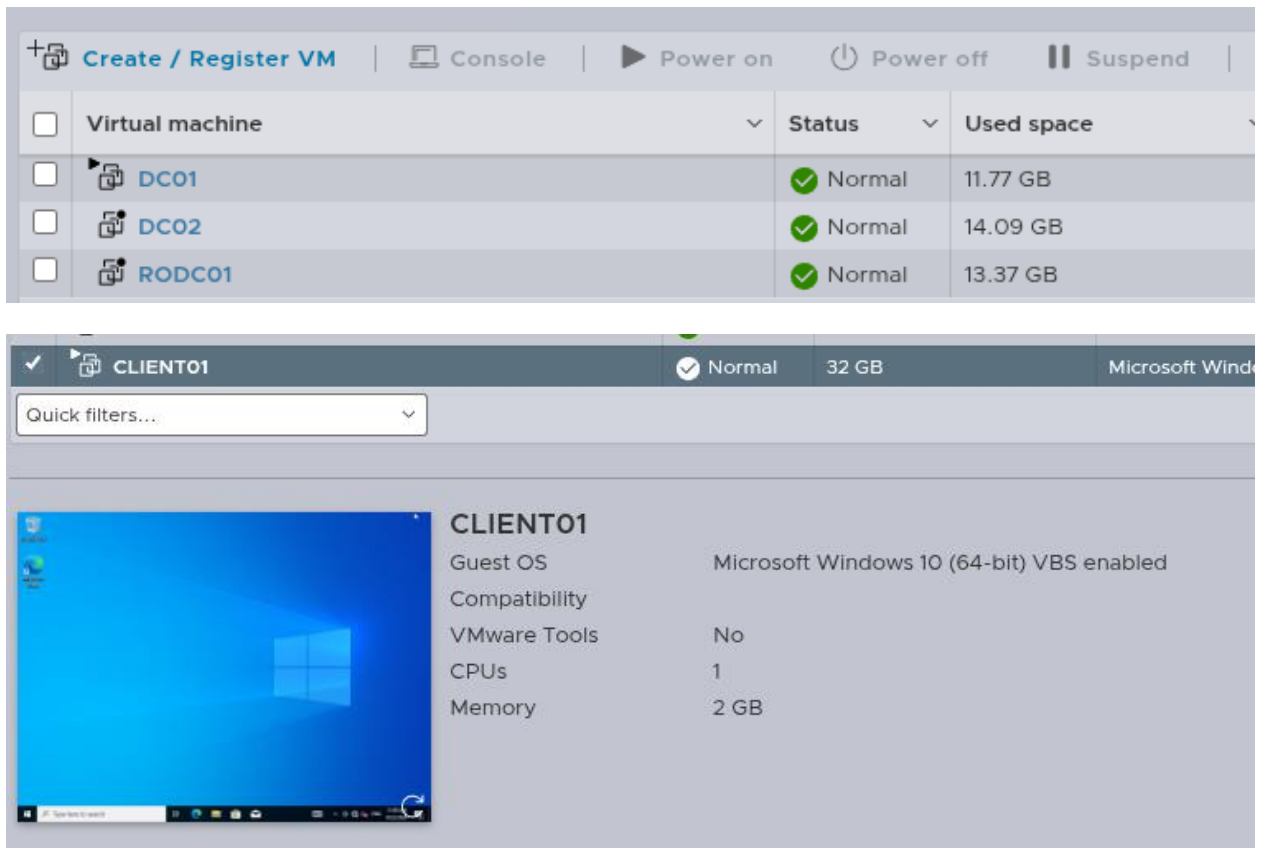
RODC01	192.168.20.10 / 24
Gateway	192.168.20.1

DCHP

Start IP	192.168.10.100
End IP	192.168.10.200
length	24
mask	255.255.255.0
Router /default gateway	192.168.10.1

Implementation & Deployment

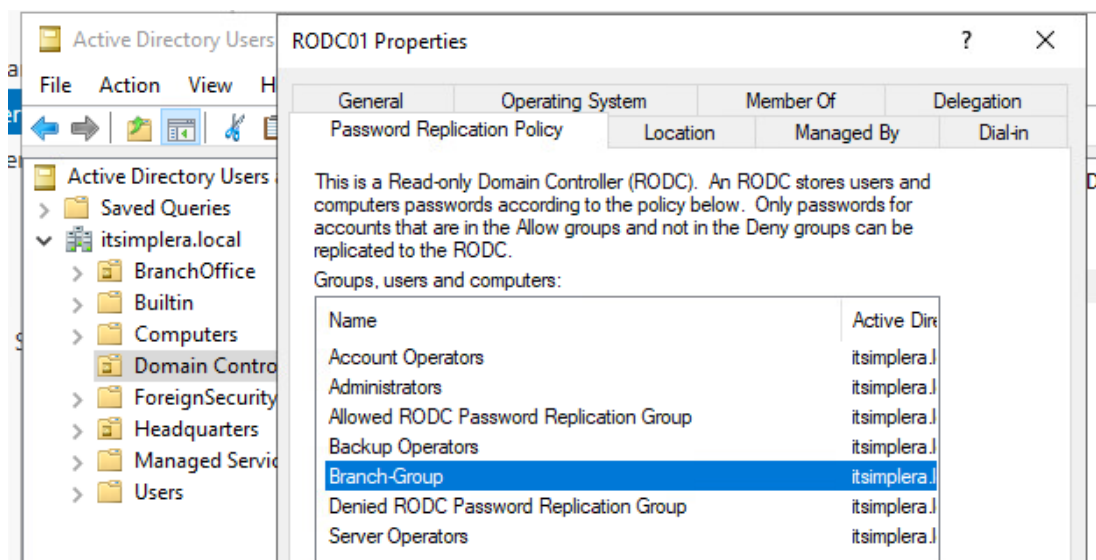
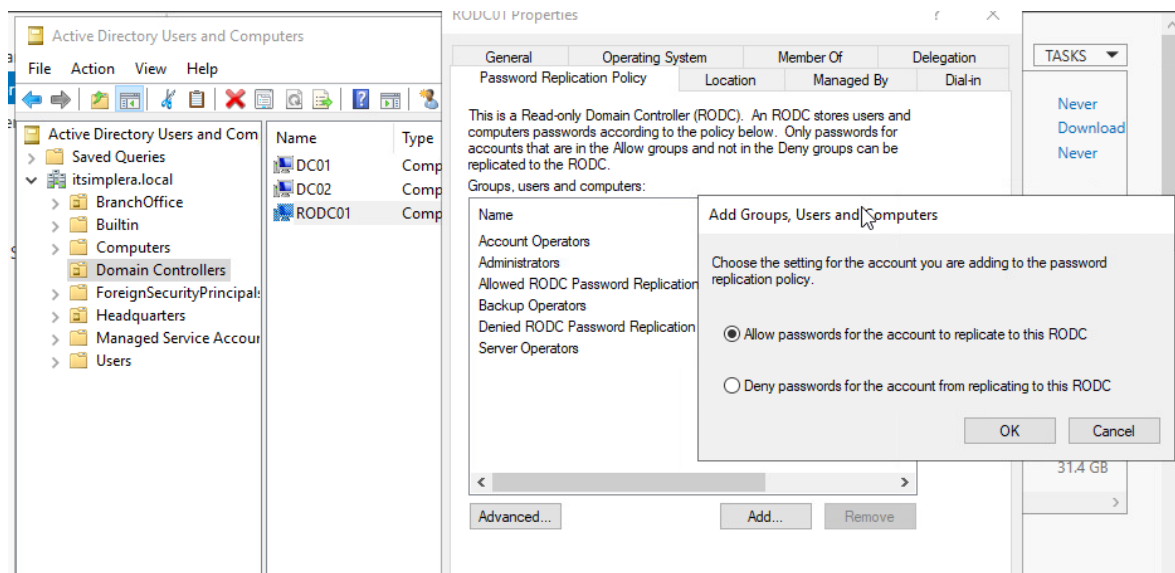
- Creating and deploying the virtual infrastructure



- Domain controllers configuration

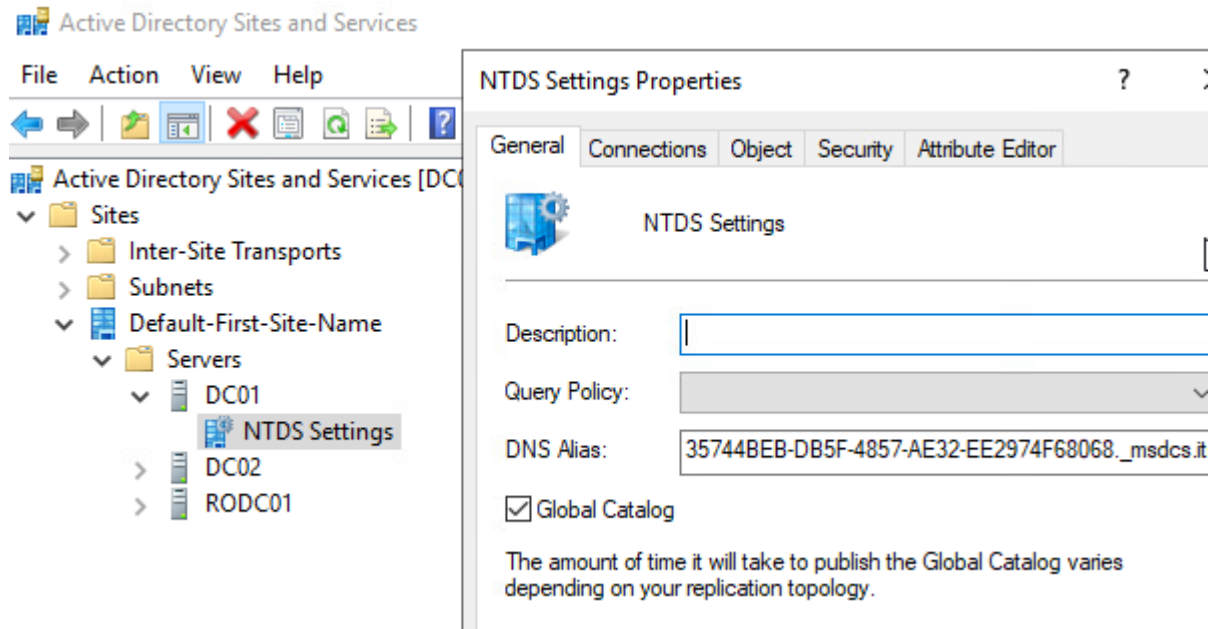
Computer name	DC01	Computer name	DC02	Computer name	RODC01
Domain	itsimplera.local	Domain	itsimplera.local	Domain	itsimplera.local
Windows Defender Firewall	Public: On	Windows Defender Firewall	Public: On	Windows Defender Firewall	Domain: On, Public: On
Remote management	Enabled	Remote management	Enabled	Remote management	Disabled
Remote Desktop	Disabled	Remote Desktop	Disabled	Remote Desktop	Disabled
NIC Teaming	Disabled	NIC Teaming	Disabled	NIC Teaming	Disabled
Ethernet0	192.168.10.10, IPv6 enabled	Ethernet0	192.168.10.11, IPv6 enabled	Ethernet0	192.168.20.10
				Ethernet1	192.168.10.12

- Configure the Password Replication Policy (PRP) to allow branch users' passwords to cache locally

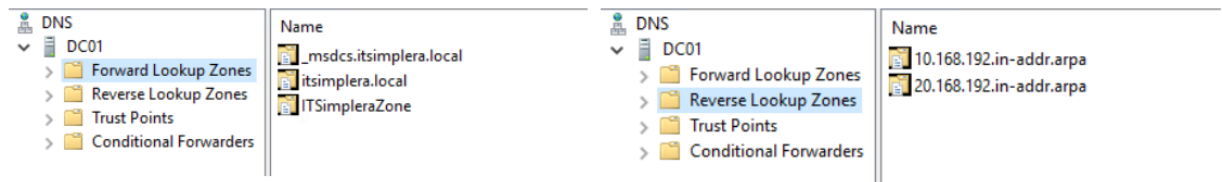


- Global catalog verification

The Global Catalog (GC) is a special role that helps with forest-wide searches and logins . By default, the first DC in a forest is a GC



- DNS verification



```
C:\Users\Administrator>nslookup 127.0.0.1
DNS request timed out.
    timeout was 2 seconds.
Server: Unknown
Address: ::1

Name:   localhost
Address: 127.0.0.1

C:\Users\Administrator>nslookup 192.168.10.10
DNS request timed out.
    timeout was 2 seconds.
Server: Unknown
Address: ::1

Name:   DC01.itsimplera.local
Address: 192.168.10.10
```

```
DC02

Select Administrator: Command Prompt

Microsoft Windows [Version 10.0.17763.3650]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator.ITSIMPLERA>nslookup 192.168.10.10
DNS request timed out.
    timeout was 2 seconds.
Server: Unknown
Address: ::1

Name:   DC01.itsimplera.local
Address: 192.168.10.10
```

- AD replication and DNS zone transfers between DC01 and DC02

```
DC02
Administrator: Command Prompt

C:\Users\Administrator\ITSIMPLERA>repadmin /showrepl

Repadmin: running command /showrepl against full DC localhost
Default-First-Site-Name\DC02
DSA Options: IS_GC
Site Options: (none)
DSA object GUID: c8ad4d03-e4e2-4c85-bbba-c469a8ef547c
DSA invocationID: 5b392055-2321-4b9c-bdec-9f9016606955

===== INBOUND NEIGHBORS =====

DC=itsimplera,DC=local
  Default-First-Site-Name\DC01 via RPC
    DSA object GUID: 35744beb-db5f-4857-ae32-ee2974f68068
    Last attempt @ 2026-08-21 23:23:23 was successful.

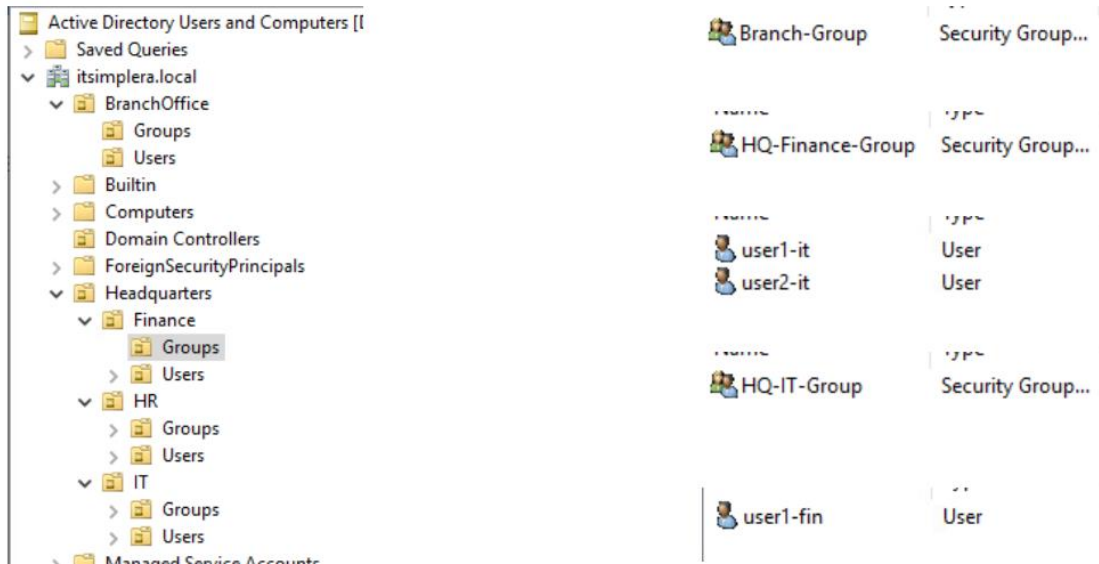
CN=Configuration,DC=itsimplera,DC=local
  Default-First-Site-Name\DC01 via RPC
    DSA object GUID: 35744beb-db5f-4857-ae32-ee2974f68068
    Last attempt @ 2026-08-21 22:56:13 was successful.

CN=Schema,CN=Configuration,DC=itsimplera,DC=local
  Default-First-Site-Name\DC01 via RPC
    DSA object GUID: 35744beb-db5f-4857-ae32-ee2974f68068
    Last attempt @ 2026-08-21 22:50:43 was successful.

DC=DomainDnsZones,DC=itsimplera,DC=local
  Default-First-Site-Name\DC01 via RPC
    DSA object GUID: 35744beb-db5f-4857-ae32-ee2974f68068
    Last attempt @ 2026-08-21 23:22:27 was successful.

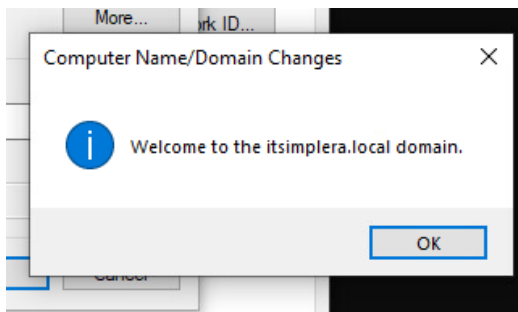
DC=ForestDnsZones,DC=itsimplera,DC=local
  Default-First-Site-Name\DC01 via RPC
    DSA object GUID: 35744beb-db5f-4857-ae32-ee2974f68068
    Last attempt @ 2026-08-21 22:50:43 was successful.
```


- OUs architecture

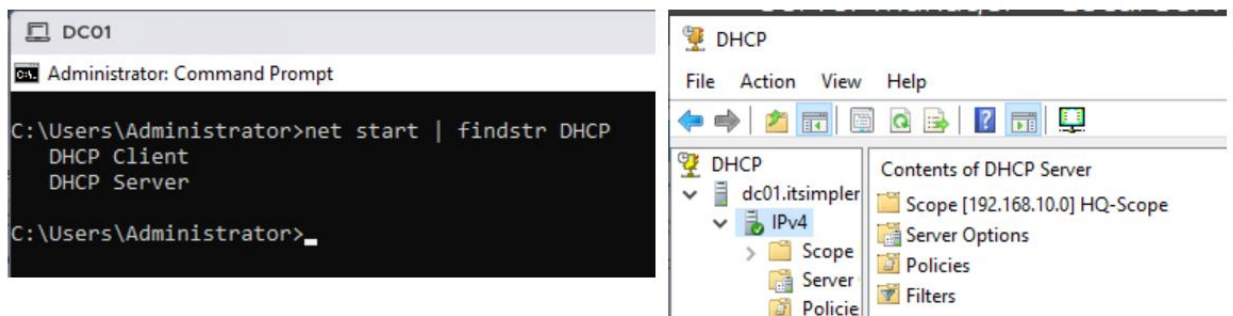


- Joining client to domain

sysdm.cpl → change → choose domain



- Verifying DHCP



on client: set preferred DNS to server IP address

```
C:\Users\u1-it>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : itsimplera.local
    IPv4 Address. . . . . : 192.168.10.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

C:\Users\u1-it>ipconfig /all

Windows IP Configuration

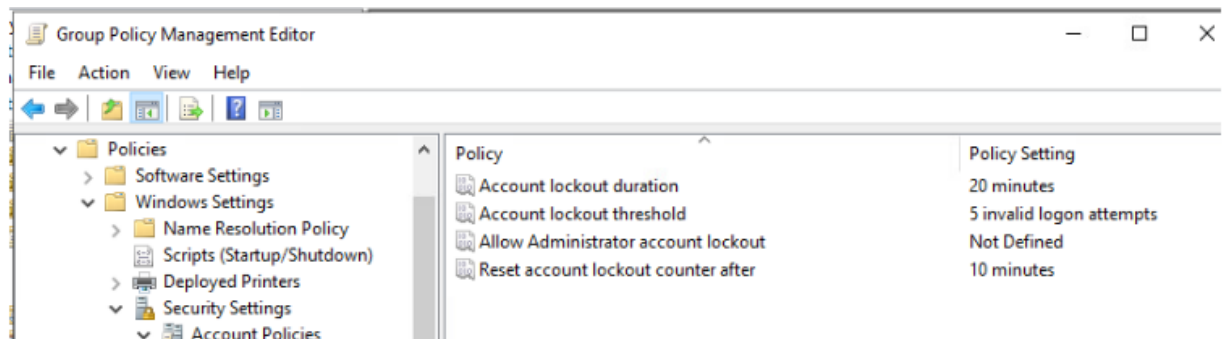
    Host Name . . . . . : DESKTOP-N5TERUB
    Primary Dns Suffix . . . . . : itsimplera.local
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : itsimplera.local

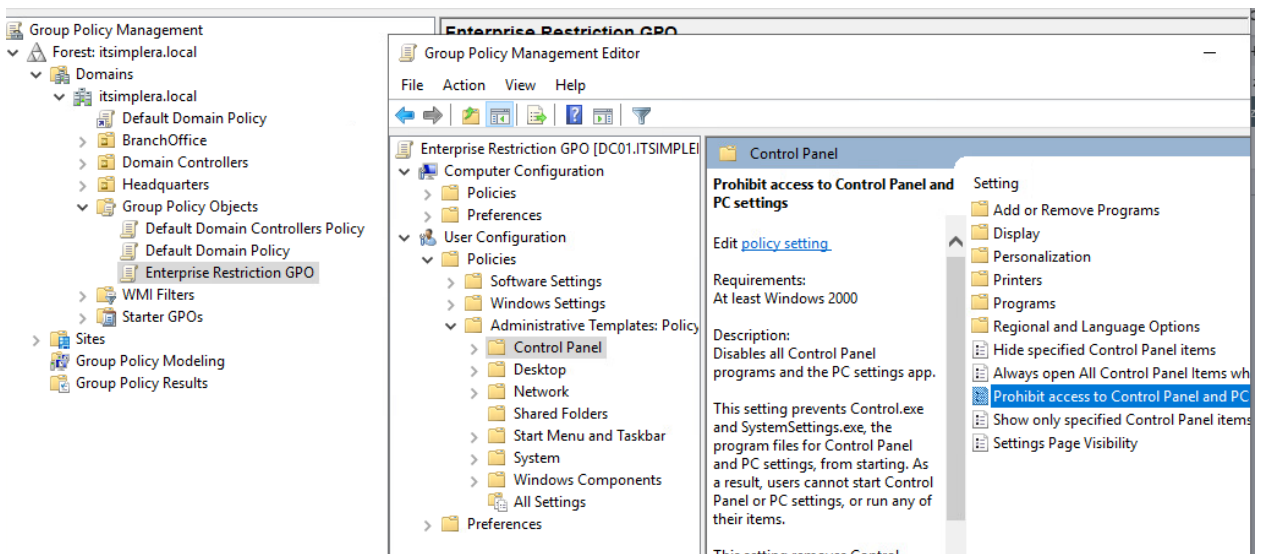
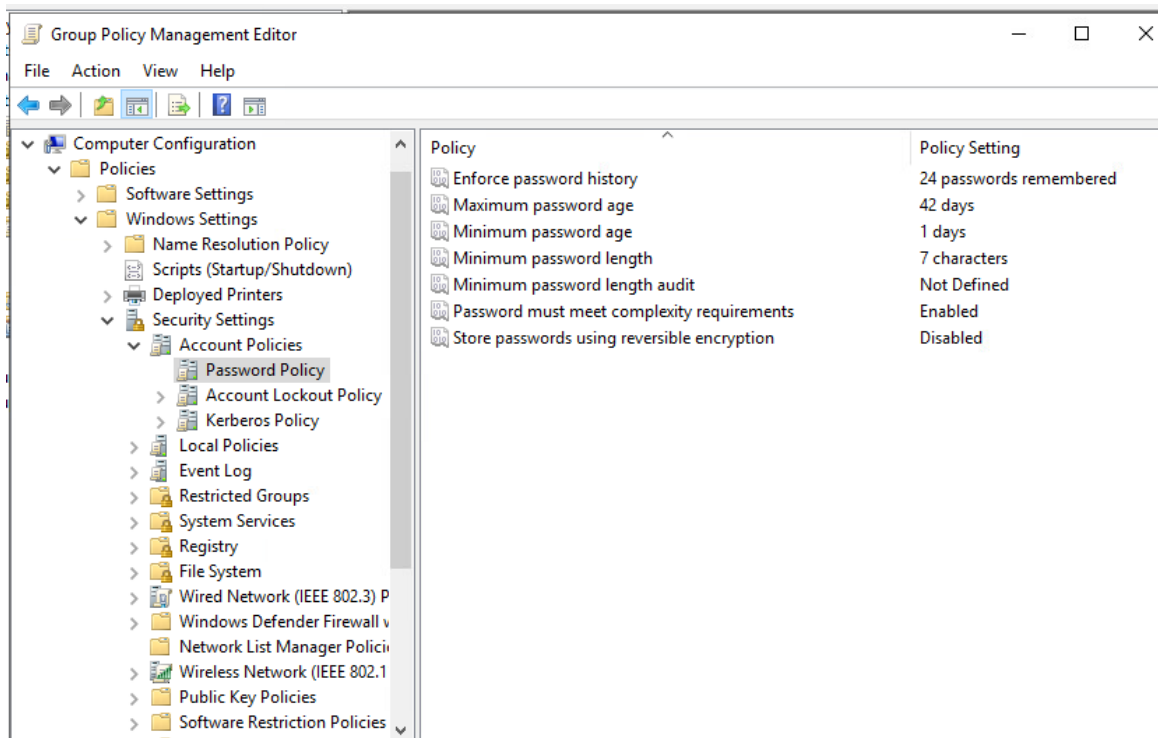
Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : itsimplera.local
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-95-B7-9A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.10.100(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Saturday, August 22, 2026 11:56:17 AM
    Lease Expires . . . . . : Sunday, August 30, 2026 11:58:51 AM
    Default Gateway . . . . . :
    DHCP Server . . . . . : 192.168.10.10
    DNS Servers . . . . . : 192.168.10.10
    NetBIOS over Tcpip. . . . . : Enabled

C:\Users\u1-it>
```

- Implementing group policies





- Confirm GPO application

```
RSOP data for ITSIMPLERA\Administrator on DC01 : Logging Mode
-----

OS Configuration:          Primary Domain Controller
OS Version:                10.0.17763
Site Name:                 Default-First-Site-Name
Roaming Profile:           N/A
Local Profile:             C:\Users\Administrator
Connected over a slow link?: No

COMPUTER SETTINGS
-----
CN=DC01,OU=Domain Controllers,DC=itsimplera,DC=local
Last time Group Policy was applied: 8/22/2026 at 12:01:17 AM
Group Policy was applied from:    DC01.itsimplera.local
Group Policy slow link threshold: 500 kbps
Domain Name:                     ITSIMPLERA
Domain Type:                     Windows 2008 or later

Applied Group Policy Objects
-----
    Default Domain Controllers Policy
    Default Domain Policy

The following GPOs were not applied because they were filtered out
-----
    Local Group Policy
        Filtering:  Not Applied (Empty)

The computer is a part of the following security groups
-----
    BUILTIN\Administrators
    Everyone
    BUILTIN\Pre-Windows 2000 Compatible Access
    BUILTIN\Users
    Windows Authorization Access Group
    NT AUTHORITY\NETWORK
    NT AUTHORITY\Authenticated Users
    This Organization
    DC01$
    Domain Controllers
    NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS
```

NT AUTHORITY\ENTERPRISE DOMAIN CONTROLLERS
Authentication authority asserted identity
Denied RODC Password Replication Group
System Mandatory Level

USER SETTINGS

CN=Administrator,CN=Users,DC=itsimplera,DC=local
Last time Group Policy was applied: 8/22/2026 at 12:03:04 AM
Group Policy was applied from: DC01.itsimplera.local
Group Policy slow link threshold: 500 kbps
Domain Name: ITSIMPLERA
Domain Type: Windows 2008 or later

Applied Group Policy Objects

N/A

The following GPOs were not applied because they were filtered out

Local Group Policy
Filtering: Not Applied (Empty)

The user is a part of the following security groups

Domain Users
Everyone
BUILTIN\Administrators
BUILTIN\Users
BUILTIN\Pre-Windows 2000 Compatible Access
NT AUTHORITY\INTERACTIVE
CONSOLE LOGON
NT AUTHORITY\Authenticated Users
This Organization
LOCAL
Group Policy Creator Owners
Domain Admins
Enterprise Admins
Schema Admins
Authentication authority asserted identity
Denied RODC Password Replication Group
High Mandatory Level

Troubleshooting

1. Not enough space for the client VM

Cause: no space available

Solution: expanding the disk for ESXi on host machine (from VMware work station, using a different hard disk)

2. RODC01 unable to join the itsimplera.local domain.

Cause: Network Isolation, RODC01 is configured with a single network adapter on the Branch Office subnet.

Solution: adding a second adapter to RODC01 on the same HQ subnet, and disabling IPv6 on both adapters.

3. CLIENT01 machine can't be assigned a dynamic IP address

Cause: incorrect DNS configuration.

Solution: setting preferred DNS to DC01 IP address.